

ZEEBRUGGE, Belgium

WMO: 064180

Lat: 51.347N

Lon: 3.202E

Elev: 30

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.6	28.7	18.2	13.7	26.8	22.3	16.8	29.5	34.7	45.7	32.5	44.9	15.0	80	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	76.6	66.6	73.3	65.0	70.9	63.8	68.0	75.5	66.3	71.8	64.9	69.7	11.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	93.1	71.9	63.8	88.6	70.1	62.5	84.8	68.8	32.4	75.7	31.0	71.6	29.9	69.9	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.4	27.9	25.2	DB	24.0	85.7	5.5	3.2	20.1	88.0	16.9	89.9	13.8	91.7	9.9	94.1	(4)
(5)			WB	23.1	71.7	5.1	1.6	19.4	72.9	16.5	73.8	13.6	74.8	9.9	75.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	40.8	40.2	44.5	50.1	55.0	60.4	64.9	64.9	61.1	55.6	48.0	42.9	(6)
(7)		DBStd	10.10	5.85	5.97	5.44	5.20	5.08	4.18	4.06	3.51	3.89	4.74	5.14	5.85	(7)
(8)		HDD50	1140	287	275	181	59	8	0	0	0	0	10	96	224	(8)
(9)		HDD65	4702	752	694	636	446	313	149	49	43	130	295	511	684	(9)
(10)		CDD50	2032	1	1	10	63	165	313	461	462	333	182	35	6	(10)
(11)		CDD65	116	0	0	0	0	4	12	45	40	13	2	0	0	(11)
(12)		CDH74	276	0	0	0	0	7	28	137	71	32	1	0	0	(12)
(13)		CDH80	59	0	0	0	0	0	4	39	11	5	0	0	0	(13)
(14)	Wind	WSAvg	13.6	15.1	14.8	13.6	12.6	13.0	13.2	12.3	12.2	12.7	13.7	14.6	15.5	(14)
(15)	Precipitation	PrecAvg	32.2	2.6	2.3	1.9	1.7	2.2	2.6	3.0	3.2	2.7	3.0	3.5	3.3	(15)
(16)		PrecMax	38.0	5.2	4.4	5.0	3.2	4.6	5.5	6.3	7.3	7.6	5.5	6.5	7.1	(16)
(17)		PrecMin	25.2	0.1	0.6	0.4	0.1	0.7	1.0	0.7	0.4	0.4	0.7	0.4	0.8	(17)
(18)		PrecStd	3.7	1.3	1.0	1.1	0.9	0.9	1.2	1.4	1.9	1.7	1.2	1.4	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	54.3	61.0	68.3	74.2	78.2	85.9	80.8	78.1	71.8	61.7	55.8	(19)
(20)			MCWB	51.3	48.4	52.1	56.4	61.9	68.1	69.4	69.7	67.2	62.6	58.4	52.8	(20)
(21)		2%	DB	51.5	51.5	56.6	64.1	68.0	71.4	76.9	75.4	72.3	67.2	59.0	54.1	(21)
(22)			MCWB	49.4	47.8	50.5	54.8	60.0	64.3	67.3	66.4	64.8	60.7	55.9	51.6	(22)
(23)		5%	DB	49.5	49.5	53.2	60.6	64.6	68.7	73.3	72.4	69.5	64.0	57.3	52.5	(23)
(24)			MCWB	47.2	46.5	48.7	53.2	58.4	62.4	65.3	65.1	63.1	59.2	54.3	50.1	(24)
(25)		10%	DB	48.0	47.6	51.1	57.0	62.1	66.1	70.6	70.2	66.8	61.9	55.2	50.6	(25)
(26)			MCWB	45.9	45.0	47.5	51.7	57.0	60.4	63.7	63.5	61.1	58.1	52.3	48.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	50.3	53.3	58.0	63.6	68.5	70.6	70.0	68.5	63.6	58.9	53.8	(27)
(28)			MCDWB	53.2	52.4	59.7	66.4	68.7	76.4	80.8	78.5	76.3	70.1	60.6	55.1	(28)
(29)		2%	WB	49.7	48.2	51.2	55.6	61.1	65.7	68.2	67.5	65.8	61.2	56.3	52.0	(29)
(30)			MCDWB	51.1	50.5	55.6	62.8	66.9	70.9	76.4	73.7	71.2	66.1	58.6	53.8	(30)
(31)		5%	WB	47.6	46.7	49.5	53.7	59.1	62.9	66.2	65.9	63.9	59.7	54.2	50.4	(31)
(32)			MCDWB	49.2	49.2	52.5	59.4	64.2	67.2	71.8	70.9	68.1	63.3	56.7	52.3	(32)
(33)		10%	WB	46.2	44.9	47.9	52.1	57.4	61.1	64.8	64.6	62.1	58.4	52.3	48.4	(33)
(34)			MCDWB	47.9	47.3	50.7	56.9	61.8	65.4	69.5	69.3	66.3	61.5	54.7	50.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	6.9	7.5	8.6	7.7	7.4	8.3	8.9	8.5	7.5	6.3	5.9	(35)
(36)			MCDBR	6.8	9.2	10.9	14.3	12.1	11.9	12.5	11.9	12.3	10.0	7.7	6.3	(36)
(37)			MCWBR	7.1	7.4	7.4	8.3	6.8	7.5	7.2	6.9	7.3	6.9	7.0	6.1	(37)
(38)		5% WB	MCDBR	7.0	8.7	10.0	12.2	10.4	10.2	11.0	10.6	10.3	9.2	7.2	6.4	(38)
(39)			MCWBR	7.5	7.3	7.6	7.8	6.6	7.2	6.9	7.0	7.0	7.0	6.9	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.354	0.390	0.416	0.425	0.421	0.424	0.415	0.393	0.371	0.347	0.330	(40)	
(41)		taud	2.372	2.357	2.262	2.227	2.253	2.283	2.284	2.325	2.373	2.411	2.382	2.348	(41)	
(42)		Ebn at Noon	213	242	253	260	264	265	262	258	250	233	205	193	(42)	
(43)		Edh at Noon	20	26	35	40	41	40	39	36	31	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	241	463	856	1404	1643	1776	1677	1433	1052	599	281	182	(44)	
(45)		RadStd	20	84	106	124	118	137	169	117	69	51	29	22	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page